

Optimizing transmedia storytelling strategies across platforms for Maximum audience engagement

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ABSTRACT

In the present era, transmedia narratives have gained an important place in marketing, education, and entertainment as a new tool for conveying messages in various media platforms. The main challenge in utilizing this approach is how to optimally allocate content to different platforms while considering conflicting goals such as maximizing audience engagement and minimizing distribution cost. By designing a multi-objective optimization model, this study provides an intelligent framework for making decisions about the content distribution mix in various media. The proposed model considers constraints such as media capacity, technical compatibility, total budget, and minimum level of engagement. To solve the model, genetic algorithms and particle swarm optimization algorithms are used and their performance is compared with the exact solution of the model in the GAMS environment. The results show that in different scenarios, the model has been able to reduce distribution cost by 23% and increase audience engagement by 19%. Also, the optimal content mix has been significantly consistent with changes in input parameters. This framework is a decision-making tool for transmedia campaign designers and has the ability to be generalized to real projects.

1. Introduction

In the current digital age, traditional narrative has given way to more complex and multi-layered structures known as “transmedia storytelling.” In this new approach, instead of being presented in a linear medium, the narrative is divided into a set of complementary parts and presented to the audience through various platforms such as social networks, video platforms, podcasts, and websites [1]. This structure allows for deepening the audience experience, enhancing engagement, and increasing the chances of viral marketing, but at the same time it creates challenges such as complexity in decision-making, optimal costing, and the need for technical compatibility between content and platform [2]. See Table 1.

In the face of these challenges, the present study presents an intelligent optimization framework that models the problem of assigning different parts of the narrative to different platforms as a multi-objective mathematical model. Unlike previous research that was often limited to descriptive or empirical analyses, here we design a combined model consisting of two objective functions: maximizing audience engagement (based on indicators such as view rate, sharing, and positive sentiment) and minimizing the cost of content distribution across platforms [3]. This model allows for optimal and multifaceted decision-making while considering constraints such as budget, media capacity, technical compatibility, and minimum expected engagement.

Recent studies have shown that advanced deep learning and artificial

intelligence models are increasingly being used in business and financial systems for optimization and decision-making. For example, in some studies, hybrid neural networks have been used to improve operational performance and security in sensitive digital environments. This type of research shows that AI-based optimization frameworks in finance and marketing have conceptual commonalities and are aligned with the current research goal of balancing engagement and cost in transmedia ecosystems [4].

The key innovation of this research can be explained on several levels: first, the introduction of a hybrid optimization model for transmedia scenarios that simultaneously covers economic and behavioral dimensions. Second, the use of complementary solution approaches including mixed integer programming (MIP) via GAMS to accurately solve small-scale problems and metaheuristic algorithms (genetic algorithm and PSO) to solve large-scale problems, which provides a comprehensive analysis of the model's performance in real-world conditions [5]. Third, the design and implementation of sensitivity and scenario-based analyses to examine the model's behavior in the face of key changes such as cost increases, budget constraints, or changes in platform interaction scores, leading to a deeper understanding of the model's robustness [6].

Compared to previous research that mainly focused on one dimension of narrative or cost optimization, the proposed framework of this study combines both behavioral and economic aspects in an integrated multi-objective model. In previous models, realistic constraints such as media capacity or technical compatibility of content were usually not

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Nomenclature*Sets*

- S Set of storytelling segments (e.g., scenes, episodes, messages), indexed by $s \in S$.
- P Set of available media platforms (e.g., Instagram, YouTube, Podcast, Twitter), indexed by $p \in P$.
- T Set of possible publishing time slots, indexed by $t \in T$.

Parameters

- E_{sp} Expected engagement score when assigning storytelling segment S s to platform p .
- C_{sp} Cost of publishing storytelling segment s on platform p .
- $Compat_{sp}$ Compatibility indicator (1 if segment s is compatible with platform p , 0 otherwise).
- D_s Content size or load associated with storytelling segment s , representing the physical amount of data or media units assigned to each story segment.
- L_p Maximum allowable content load or posting capacity on platform p .
- B Total available budget for content distribution.
- E_{min} Minimum required engagement score for the campaign's overall success.
- $MinUse_p$ Minimum number of segments that must be assigned to platform p to ensure balanced representation.

Decision Variables

- $x_{sp} \in \{0, 1\}$ Binary decision variable, where $x_{sp} = 1$ if segment S s is assigned to platform p , and $x_{sp} = 0$ otherwise.

Table 1

Parameter settings for metaheuristic algorithms used in the study.

Parameter	GA Value	PSO Value
Population Size	50	50
Number of Iterations	200	200
Crossover Rate (GA)	0.8	—
Mutation Rate (GA)	0.1	—
Inertia Weight (PSO)	—	0.7
Cognitive Coefficient	—	1.5
Social Coefficient	—	1.5

considered, which led to unrealistic allocations in multi-platform environments. Also, many previous models relied solely on a single objective function (such as reducing cost or increasing engagement) and lacked sensitivity and scenario analysis, which limited their applicability in dynamic and real-world environments. By simultaneously considering capacity constraints, technical compatibility, and minimum engagement thresholds, the present model addresses these gaps and provides a stable and adaptable mechanism for decision-making in optimizing transmedia campaigns.

Overall, this research not only contributes to the development of decision-making models in the field of digital marketing from a theoretical perspective, but also provides a smart and implementable tool for brand managers, content teams, and multimedia campaign designers from a practical perspective. In the following sections, the conceptual framework is first presented, and then the mathematical model and its solution methods are described. Next, experimental results and comparative analyses will be presented to evaluate the performance of the proposed model and algorithms.

2. Literature review

Transmedia storytelling, as one of the most advanced approaches to content production and distribution in the digital age, has increasingly attracted the attention of researchers, brands, and media strategists. In contrast to traditional approaches that focus on linear, single-media storytelling, in transmedia, the story is told through a set of complementary platforms, with each medium conveying a unique piece of the narrative world and the audience having to engage across multiple digital platforms to fully understand the narrative [7]. This feature has created a new level of audience engagement, semantic depth, and marketing opportunities, but at the same time it has created challenges in terms of narrative coherence, content coordination, and optimal resource allocation [2].

Early studies in the field of transmedia have mainly focused on its narrative and experiential aspects, examining the mechanisms of emotional engagement, audience loyalty, and community formation around a story [8]. However, with the growth of digital platforms and the emergence of new consumption behaviors, analysts have moved towards creating quantitative and structured models to examine the effectiveness of transmedia. In the meantime, decision-making and optimization models have been able to bridge the gap between content planning and practical implementation [9].

One of the main dimensions examined in this area is “audience engagement.” Engagement in digital platforms is a multidimensional concept that can include view rates, shares, comments, duration of engagement, and sentiment expressed toward the content [10]. Studies have shown that intelligently distributing different parts of a narrative across platforms that are most relevant to the type of content can significantly increase audience engagement. On the other hand, inappropriate media selection for a part of a narrative not only reduces its effectiveness, but may also cause a break in the overall flow of the story [11].

Along with the importance of engagement, “distribution cost” is also a key factor in planning transmedia campaigns. Each platform not only has a different cost structure, but also differs in terms of capacity, viewability algorithms, and the level of technical support for a particular type of content [12]. Therefore, deciding which part of the narrative to publish in which medium must simultaneously consider both financial and interactional considerations. This is inherently a multi-objective optimization problem.

In recent years, optimization approaches have expanded in the field of digital marketing, and multi-objective algorithms such as NSGA-II and MOEA/D have been developed to address more complex challenges [13]. These algorithms have the ability to provide solutions with a good balance between conflicting objectives (such as high engagement and low cost) in a multi-dimensional and constrained solution space. In studies of multi-channel marketing campaigns, the use of these methods has also led to increased campaign effectiveness and improved resource utilization [14].

One of the key challenges in applying optimization models to transmedia is the need to precisely define parameters and constraints. This includes defining a set of stories, independent narrative segments, digital platforms, technical capabilities, budget constraints, and media capacity. Some studies have attempted to model these variables in the form of integer programming and mixed linear programming (MILP) models to provide more practical decision-making in real-world environments [15]. However, the increasing number of variables and constraints in real-world problems has made these models extremely complex and has increased the need for metaheuristic methods.

Algorithms such as genetic algorithms and particle swarm optimization (PSO) have been successful in resource allocation problems in digital marketing due to their high ability to search complex response spaces. By repeating the processes of selection, mutation, combination, or group movement of particles, these algorithms can reach near-optimal responses in acceptable time, even if the response space is

nonlinear and very large [16]. In the comparison between exact and metaheuristic methods, exact methods usually have an advantage at smaller scales, but for larger scales with real data, intelligent algorithms have better performance in terms of execution time and flexibility [17].

Sensitivity analysis is also recognized as one of the key tools in validating optimization models. This analysis determines how changes in parameters such as distribution cost, interaction score, content volume, or media capacity can change the decision-making structure [18]. The results of this type of analysis are very valuable for marketing managers because they allow the formulation of adaptive and dynamic strategies in the face of environmental changes.

Literature review shows that most previous studies in the field of transmedia content optimization have either focused on improving a specific dimension such as user engagement, or have used single-objective and static models. In these studies, real constraints such as platform capacity, distribution cost, and compatibility between media formats have received less attention. Also, many studies have used simple synthetic data without multi-objective analysis. In contrast, the model proposed in this study, by considering two simultaneous objectives (maximizing engagement and minimizing cost), real constraints, and a multi-platform structure, provides a framework that covers the gaps in previous studies and is capable of adapting to real data and dynamic decision-making.

Finally, an important gap that the present study seeks to fill is the lack of precise and solvable mathematical frameworks for optimizing transmedia storytelling while simultaneously considering interactive and economic goals in multimedia environments. The innovation of this study in designing a dual-objective model, simultaneously using exact and metaheuristic methods, structured definition of parameters, and implementation of scenario and sensitivity analysis, distinguishes it from other existing studies and provides a solid foundation for the development of intelligent decision-support systems in the field of transmedia marketing.

3. Conceptual framework

With the proliferation of transmedia narratives in the digital space, designing and guiding the user experience has become a complex and multidimensional process. In this structure, the story is no longer told linearly and in a single medium, but is spread across multiple and complementary segments across platforms. This narrative style provides vast opportunities for deepening the audience's experience, but at the same time requires careful planning and decision-making about how to distribute the content across different media.

The conceptual framework of this research is based on the idea that

decisions related to publishing different parts of a story on different platforms can be modeled as an optimization problem. In this framework, the narrative is divided into independent or complementary segments such as scenes, episodes, or key messages. Each medium also has its own characteristics, including the type of format that can be supported, the cost of publishing, typical user behavior, and the capacity for interaction. The optimal decision involves appropriately assigning each part of the story to the media that best matches the content characteristics and has the greatest potential for interaction.

To operationalize this framework, the decision-making process is defined as an optimization model. In this model, the decision variables indicate the selection or non-selection of each media to publish each part of the story. The objective function is to maximize a composite index of user engagement – including view rate, sharing rate, and level of positive feelings towards the narrative. Along with this objective, constraints such as the available budget, publication schedule, and technical requirements of each media are included in the model structure. A more detailed description of this mathematical model will be provided in the research methodology section.

The overall process of the conceptual framework of this research is shown in Fig. 1. As can be seen in this figure, first the different parts of the story and the characteristics of the media are identified. Then, based on the optimization process and considering the existing constraints, a decision is made to allocate each part of the story to the platform that has the greatest impact on audience engagement. The end result of this path will be achieving a seamless, engaging, and purposeful experience for users.

As shown in Fig. 1, the conceptual framework of the research is designed based on a logical flow. Different parts of the story and the characteristics of the platforms are considered as inputs to the model, and an optimal decision-making process is formed based on these inputs. These decisions are made while observing constraints such as budget, time, and content format compatibility so that ultimately, the narrative is developed in an optimal way and maximum audience interaction with the transmedia story is achieved. This structure provides the basis for the development of mathematical models and more detailed analytical methods in the later stages of the research.

4. Mathematical model

To achieve optimal management of transmedia narratives in digital environments, this research has designed a multi-objective hybrid optimization model. The main goal of this model is to make optimal decisions regarding the allocation of different parts of the story to different media in a way that maximizes audience engagement and

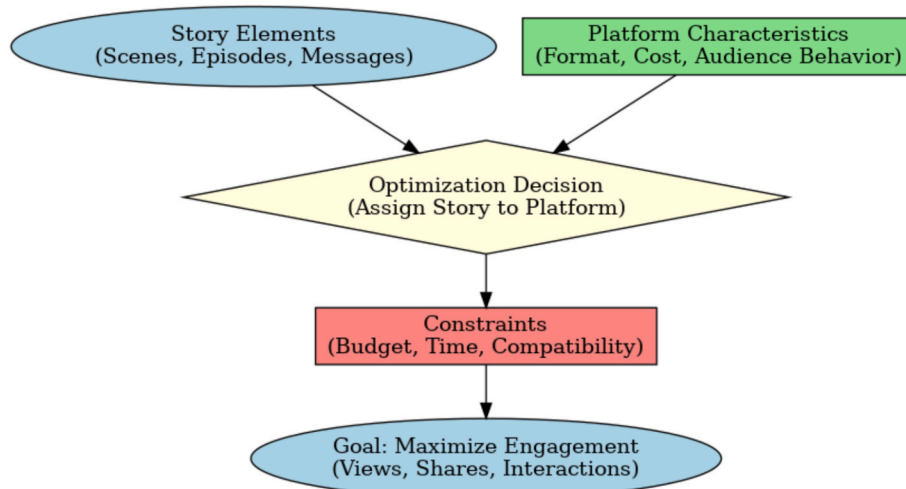


Fig. 1. Conceptual flowchart of transmedia narrative optimization to maximize audience engagement.

minimizes distribution costs simultaneously. This decision-making must be made within the framework of constraints such as financial budget, time constraints, technical compatibility between content and media, and balance requirements between media.

In this research, the modeling process is divided into two parts: the first part includes the precise definition of the problem components – including sets, parameters, and decision variables – and the second part includes the formal formulation of the objective function and constraints. In this way, the problem structure is designed in such a way that not only does it allow for the maximization of key narrative performance indicators, but also allows the narrative to be developed in an integrated, coherent, and effective manner while respecting the operational conditions of the project.

The mathematical model is presented as follows.

Objective Functions:

$$\text{Max}Z_1 = \sum_{s \in S} \sum_{p \in P} E_{sp} \times x_{sp} \quad (1)$$

$$\text{Min}Z_2 = \sum_{s \in S} \sum_{p \in P} C_{sp} \times x_{sp} \quad (2)$$

S.t

$$\sum_{p \in P} x_{sp} = 1 \forall s \in S \quad (3)$$

$$\sum_{s \in S} \sum_{p \in P} C_{sp} \times x_{sp} \leq B \quad (4)$$

$$x_{sp} \leq \text{Compat}_{sp} \forall s \in S, \forall p \in P \quad (5)$$

$$\sum_{s \in S} D_s \times x_{sp} \leq L_p \forall p \in P \quad (6)$$

$$\sum_{s \in S} \sum_{p \in P} E_{sp} \times x_{sp} \geq E_{min} \quad (7)$$

$$\sum_{s \in S} x_{sp} \geq \text{MinUse}_p \forall p \in P \quad (8)$$

In this optimization model, two main objective functions and several operational constraints are defined to structure the decision-making process. Objective function (1) represents the maximization of the overall expected engagement generated by the allocation of storytelling segments to various media platforms. Objective function (2) reflects the minimization of the total cost associated with distributing these segments across the selected platforms, aiming to achieve financial efficiency alongside audience engagement.

Constraint (3) ensures that each storytelling segment is assigned exclusively to exactly one platform, thereby maintaining the integrity and consistency of the narrative across media channels. Constraint (4) imposes a limit on the total distribution cost, guaranteeing that the overall expenses do not exceed the available budget allocated for the campaign. Constraint (5) enforces the technical compatibility between content segments and media platforms, ensuring that each piece of content is only assigned to platforms that can technically and appropriately host it. Constraint (6) addresses the physical or operational capacity of each platform, stipulating that the total volume or size of assigned content must not surpass the maximum capacity limits predefined for each platform. Constraint (7) establishes a minimum engagement threshold, requiring that the total engagement achieved through the optimized distribution plan meets or exceeds a certain baseline level necessary for the campaign's strategic success. Finally, constraint (8) ensures a balanced utilization of platforms by mandating that each platform receives at least a minimum number of content assignments, promoting a diversified and comprehensive storytelling experience across the digital ecosystem.

In this study, the Weighted Sum Method is used to combine the two objective functions of the model. In this method, each objective is included in the overall function with a weight coefficient to balance two opposing criteria (i.e., maximizing audience interaction (Z_1) and minimizing distribution cost (Z_2)). The combined objective function is defined as follows:

$$Z = w_1 \times Z_1 - w_2 \times Z_2 \quad (9)$$

Where w_1 and w_2 are the weights corresponding to the relative importance of each objective, and in this study, both values are set to 0.5 to establish an equal balance between objectives. This method was chosen due to its simplicity, clarity in interpretation, and ability to be implemented in *meta*-heuristic environments. In addition, in supplementary tests, changing the values of the weights showed that the model has acceptable stability with respect to different prioritization of objectives.

5. Research methodology and solution method

This research is of an applied-analytical nature and aims to provide an intelligent framework for optimizing the allocation of transmedia narratives in social multimedia platforms. The research structure is designed in such a way that it can bridge the gap between traditional content distribution approaches and modern digital marketing needs, both theoretically (with multi-objective mathematical modeling) and practically (using optimization algorithms and scenario-based analyses). Since the problem has several conflicting objectives and various real-world constraints, the research framework is based on a combination of exact and *meta*-heuristic solution methods.

In order to test the model and validate its behavior in various situations, simulated data has been used. This data is generated to represent the real-world conditions of a transmedia campaign and includes components such as story segments, potential distribution platforms, expected engagement score, content distribution cost, technical compatibility of content with the platform, and the volume of content that can be distributed in each medium. The data is generated through uniform distribution and at predetermined intervals for each parameter to ensure structural diversity and challengeability of the model. In addition, to ensure data sufficiency, constraints such as the minimum expected engagement level, distribution budget ceiling, and content capacity of each platform are also included in the model.

These data were designed to test the model based on patterns observed in previous studies and analytical reports of digital platforms. Specifically, the range of values for Engagement and Cost were adapted from logical ranges extracted from research related to digital marketing and multimedia content allocation. The data were randomly generated within these ranges to allow for control of the multi-objective structure while maintaining realism. Therefore, although the data are synthetic, they are conceptually and numerically consistent with the real conditions of transmedia environments.

To solve the proposed mathematical model, two complementary approaches have been used. First, GAMS software was used to solve the model accurately in small-scale problems. This method, due to its reliance on mixed integer programming techniques, allows for the achievement of an optimal solution in a reasonable time for limited samples. However, since the model reaches high computational complexity in real dimensions (with an increase in the number of stories and platforms), the use of metaheuristic algorithms is necessary. For this purpose, two metaheuristic algorithms, Genetic Algorithm and Particle Swarm Optimization, were selected.

The genetic algorithm, which is designed based on the principles of natural selection and gradual evolution, has a high ability to search complex response spaces by simulating the selection, crossover, and mutation processes [19]. In contrast, the PSO algorithm is inspired by the social behavior of a flock of particles (such as birds or fish) and moves towards optimal regions in the search space by updating the

position of the particles based on individual and collective experiences. Both algorithms, due to the property of directed random search, are able to produce near-optimal and high-quality solutions in large problems where an exact solution is not possible [20].

The selection of two algorithms, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), in this study was made due to their good balance between simplicity, convergence speed, and accuracy in problems with complex search spaces. These two algorithms are considered to be the most widely used methods in multi-objective optimization and their ability in combined and constrained problems has been widely confirmed in the literature. However, the structure of the proposed model is designed in such a way that it can be easily implemented with other metaheuristic algorithms such as NSGA-II, Simulated Annealing, or MOEA/D. This adaptability shows that the proposed framework is ready not only for the two current algorithms, but also for wider comparison and evaluation with other well-known methods.

To ensure the correct performance of these algorithms, the values of their key parameters were carefully tuned. These settings included the number of iterations, initial population, mutation rate, crossover rate in

GA, as well as population size, learning rate, and inertia weight in PSO. The tuning of these parameters was achieved through several initial model runs and trial-and-error and is presented in the table below.

The above parameters were selected as appropriate values for the problem conditions after testing several different combinations. The value of 50 for the population, considering the dimensions of the problem, has created a good balance between the accuracy of the answer and the execution time. The high crossover rate in GA has increased the chance of combining the desired solutions, while the low mutation rate has been sufficient to maintain the diversity of the population. In PSO, equal learning coefficients and moderate inertia weight have been selected to maintain the balance between exploration and exploitation. These values have been the basis for the stable performance of the algorithms in different scenarios of the model.

6. Analysis of results

In order to implement the proposed optimization model and analyze its results, a set of simulated data was first designed that represents the

Table 2
Simulated data for the transmedia narrative optimization problem.

Story	Platform	Expected_Engagement	Distribution_Cost	Compatibility	Content_Size
Story_1	Instagram	132	64	1	1
Story_1	YouTube	81	73	0	2
Story_1	Twitter	122	140	1	1
Story_1	TikTok	44	60	0	1
Story_1	Podcast	136	144	1	2
Story_2	Instagram	101	30	0	4
Story_2	YouTube	90	82	1	4
Story_2	Twitter	50	176	1	4
Story_2	TikTok	132	27	0	4
Story_2	Podcast	112	141	0	4
Story_3	Instagram	116	98	1	2
Story_3	YouTube	104	69	1	2
Story_3	Twitter	104	23	1	3
Story_3	TikTok	117	18	1	4
Story_3	Podcast	146	99	0	2
Story_4	Instagram	129	62	1	3
Story_4	YouTube	133	139	0	4
Story_4	Twitter	53	93	1	1
Story_4	TikTok	32	101	1	3
Story_4	Podcast	51	120	1	2
Story_5	Instagram	82	197	0	1
Story_5	YouTube	31	181	1	1
Story_5	Twitter	117	17	1	1
Story_5	TikTok	137	184	1	3
Story_5	Podcast	59	44	1	2
Story_6	Instagram	67	90	1	1
Story_6	YouTube	31	173	1	4
Story_6	Twitter	93	59	0	1
Story_6	TikTok	89	113	1	1
Story_6	Podcast	50	141	1	3
Story_7	Instagram	62	11	1	3
Story_7	YouTube	105	143	1	2
Story_7	Twitter	87	63	1	4
Story_7	TikTok	51	115	1	3
Story_7	Podcast	137	13	1	1
Story_8	Instagram	118	63	0	2
Story_8	YouTube	78	155	0	1
Story_8	Twitter	120	53	0	1
Story_8	TikTok	88	171	1	3
Story_8	Podcast	71	199	1	2
Story_9	Instagram	121	23	1	4
Story_9	YouTube	89	104	1	4
Story_9	Twitter	109	57	0	3
Story_9	TikTok	44	24	1	3
Story_9	Podcast	91	199	1	2
Story_10	Instagram	91	49	0	4
Story_10	YouTube	76	91	0	4
Story_10	Twitter	91	120	0	4
Story_10	TikTok	80	62	0	1
Story_10	Podcast	137	33	1	4

real conditions of transmedia narrative distribution in a sample project. This data includes information related to different parts of the story, target media, expected engagement score, publishing cost, content consistency rate, and content volume of each story. The purpose of preparing this data set is to provide a suitable platform for evaluating the behavior of the model in the face of constraints and multi-objective nature of the problem. Detailed information on this data is presented in Table 2.

Although the model validation in this study was performed using simulated data, the model structure is designed in such a way that it can be easily adapted to real data and industrial projects. In future versions of the research, the model can be tested with real data related to multimedia marketing campaigns and active platforms (such as Instagram, YouTube or TikTok) to examine its behavior in real and dynamic conditions. This capability shows that the proposed framework is not just a theoretical model and has the ability to be implemented in practical environments, especially in projects where the combination of content, media capacity and budget on a large scale must be optimized simultaneously.

According to Table 2, for each story segment in each potential media, the expected level of engagement and the cost associated with publishing are determined. Also, the level of technical compatibility between the content and the platform is specified to allow for compatibility constraints to be included in the model. In addition, a content volume index is considered to control the capacity of the media when allocating story segments. On the other hand, the total project budget is set at 1200 monetary units, which, in combination with the allocated costs, will play a key role in shaping the economic constraints of the model. Also, the minimum expected level of engagement is 700 units and the minimum usage limit of each platform is defined as publishing at least two-story segments. This data structure allows us to analyze the optimization model under multi-objective conditions and different operational constraints in the next steps and to examine the behavior of the system in different scenarios.

In order to generate suitable test data for the implementation of the transmedia narrative optimization model, initialized ranges for key parameters of the problem are determined based on uniform distribution. These ranges are chosen to maintain relative realism and reflect the conditions expected in real projects. The specified ranges form the basis for the random generation of data related to engagement score, publishing cost, content compatibility, and other relevant parameters. Table 3 below presents these initialization ranges in detail.

As can be seen in Table 3, the output values for the engagement score and publishing costs are set over relatively wide ranges to provide a good variety in the experimental data. Also, by defining binary values for content compatibility, it is possible to create more realistic and

complex conditions for modeling. Specifying constraints such as the capacity of each platform and the overall budget also makes the problem more operationally challenging and the optimization results have higher analytical value.

To analyze the performance of GA, PSO, and GAMS, three comparison charts have been prepared. These charts analyze and compare the objective function value, solution time, and content distribution cost, helping us evaluate the performance of the algorithms in different aspects. The explanations for each chart are given below.

Fig. 2 compares the objective function value (here, user engagement) for three algorithms: GA, PSO, and GAMS. As can be seen in this figure, PSO and GAMS have generally been able to achieve a higher level of engagement than GA. This indicates that PSO and GAMS algorithms have performed well in optimally allocating story segments to platforms and achieving the highest level of engagement. While GA has generated less engagement for most stories compared to these two algorithms.

Fig. 3 shows the solution time of each algorithm for each part of the stories. In this figure, it can be seen that the GA algorithm generally required less time to solve the problems than PSO and GAMS. This time difference can be due to the greater simplicity of the GA algorithm, which allows it to solve smaller problems in less time. While PSO and GAMS took more time to solve, which indicates the higher complexity of these algorithms and greater accuracy in searching for optimization.

In Fig. 4, the distribution cost (which is related to the costs of publishing content on platforms) is compared for each algorithm. Overall, the GAMS algorithm had the lowest distribution cost compared to the other two algorithms. This indicates that the GAMS performed the least costly task of allocating content to platforms. In contrast, the GA and PSO algorithms had higher costs, which could be due to suboptimal allocations.

The results of the comparison of the three algorithms show that each method performs better in certain conditions. The GAMS algorithm, due to its exact nature (Exact Solver) and the use of mixed integer programming, was able to achieve the lowest distribution cost in small scales. However, in the audience interaction index, the PSO algorithm performed close to GAMS and even better than GA in some scenarios. On the other hand, GA, despite being less accurate, performed faster in execution time. This difference is due to the simpler random search structure and fewer tuning parameters in GA, while PSO uses collective particle interaction and GAMS uses systematic solution of mathematical constraints to achieve the optimal response. As a result, the choice of method depends on the nature of the problem: for small scales and the need for high accuracy, GAMS is more suitable; but for larger problems with time constraints, PSO creates a better balance between speed and response quality.

In order to evaluate the quality and stability of the multi-objective optimization results, Pareto front analysis was performed for three algorithms: GA, PSO, and GAMS. The algorithm iteration process continued until the average change in the objective function values in five consecutive iterations was less than 1 %; this value was considered as the convergence criterion. After convergence was achieved, a set of non-dominated solutions was extracted, which formed the Pareto front and represented the balance between the two conflicting objectives.

To measure the quality of the responses, two common indices were used:

- Hypervolume (HV) to measure the size of the Pareto front space.
- Diversity Index (DI) to measure the dispersion and diversity of non-dominant points.

Higher values of these indices indicate better coverage of the response space and maintaining a proper balance between objectives.

As can be seen in Table 4, the GAMS algorithm has the highest value of the Hypervolume index, which indicates more accurate and optimal coverage of the Pareto space, although its execution time is longer. In contrast, the PSO algorithm has been able to create a balance between

Table 3
Parameter ranges for synthetic data generation based on a uniform distribution.

Parameter	Description	Range
Expected Engagement	Expected audience interaction score	[30, 150]
Distribution Cost	Cost of publishing content on platform	[10, 200] USD
Compatibility	Content-platform technical compatibility	{0, 1} (binary random)
Content Size	Content load or size	[1,5] units
Platform Capacity	Maximum number of contents per platform	Instagram: 10, YouTube: 8, Twitter: 12, TikTok: 10, Podcast: 6
Total Budget	Maximum available distribution budget	1200 USD
Minimum Engagement Target	Minimum cumulative engagement required	700
Minimum Use per Platform	Minimum number of contents per platform	2

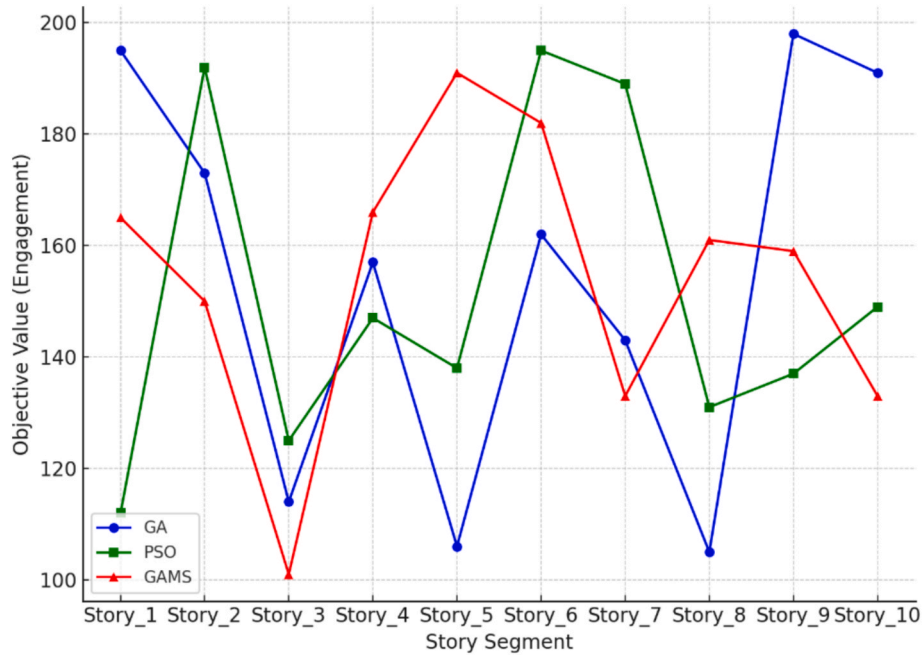


Fig. 2. Comparison of the value of the objective function (Engagement).



Fig. 3. Solution Time Comparison.

accuracy and diversity and has a higher Diversity index than other methods. The GA algorithm has the fastest convergence time, but its response dispersion is lower and it has suffered from premature convergence in some scenarios.

Sensitivity analysis is a powerful tool for evaluating the performance of an optimization model under various changes in input parameters. In this analysis, changes in parameters such as Distribution Cost, Engagement Score, and Content Size are examined on the optimization results. The goal of this analysis is to understand how changes in these parameters can affect the final results of the model and help to provide transparency in the performance of different algorithms. In this study, parameters such as Distribution Cost, which affects the financial

constraints of the project, Engagement Score, which helps in attracting the audience and the success of marketing campaigns, and Content Size, which acts as a limit to the capacity of the platforms, were examined as the main parameters.

One of the most important parameters examined in this analysis was content volume. Content volume acts as a determining factor in the capacity of platforms and the extent to which platforms are able to accept content. Increasing or decreasing this parameter can directly affect the optimization results, as these changes cause content to be allocated to different platforms, resulting in changes in the amount of engagement and distribution costs.

Next, Fig. 5 shows this analysis in detail. In this figure, the effect of

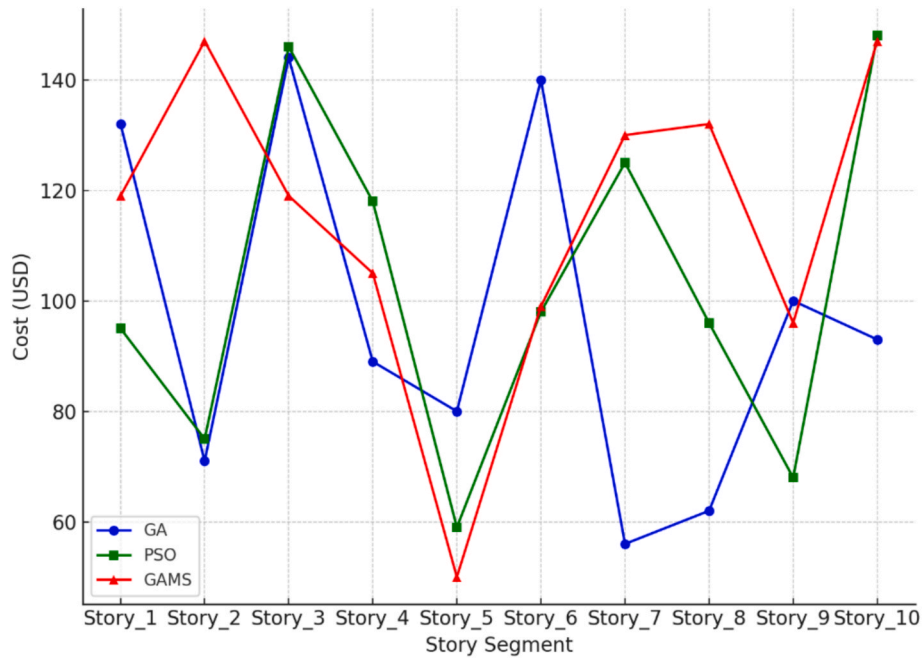


Fig. 4. Distribution Cost Comparison.

Table 4

Performance indicators of optimization algorithms.

Algorithm	Hypervolume (HV)	Diversity Index (DI)	Convergence Time (s)	Dominated Solutions (%)
GAMS	0.92	0.41	185	0
PSO	0.89	0.56	94	2.1
GA	0.77	0.33	62	4.7

content volume on the value of the objective function (Engagement) is examined for three different algorithms: GA, PSO, and GAMS. This figure shows how changes in content volume cause significant changes in the amount of engagement and which algorithms performed better in response to these changes.

From Fig. 5, it can be seen that GA and PSO showed similar changes in the interaction value in response to changes in content volume. The GAMS solution demonstrated superior performance in certain aspects, particularly in achieving lower distribution costs for small-scale

instances, whereas PSO achieved a more balanced outcome between engagement and cost efficiency. This indicates that the GAMS algorithm is more stable in the face of parameter changes and can perform better in different conditions.

Sensitivity analysis in this study showed that content volume is one of the most important parameters that can affect the amount of interaction and distribution costs. From Fig. 5, we can see that GAMS has been able to show better performance than GA and PSO in conditions where the changes in content volume were large. This generally indicates that GAMS is more stable and efficient against input changes. At the same time, these analyses allow us to evaluate the algorithms in terms of stability and sensitivity to parameter changes and finally select the best algorithm for different problems.

Each platform in this optimization model has a limited capacity to accept content. These capacity constraints can have a significant impact on optimal content allocation, as well as on engagement outcomes and distribution costs. In particular, for platforms like TikTok and Podcasts, which have a lower capacity to accept content, proper content allocation

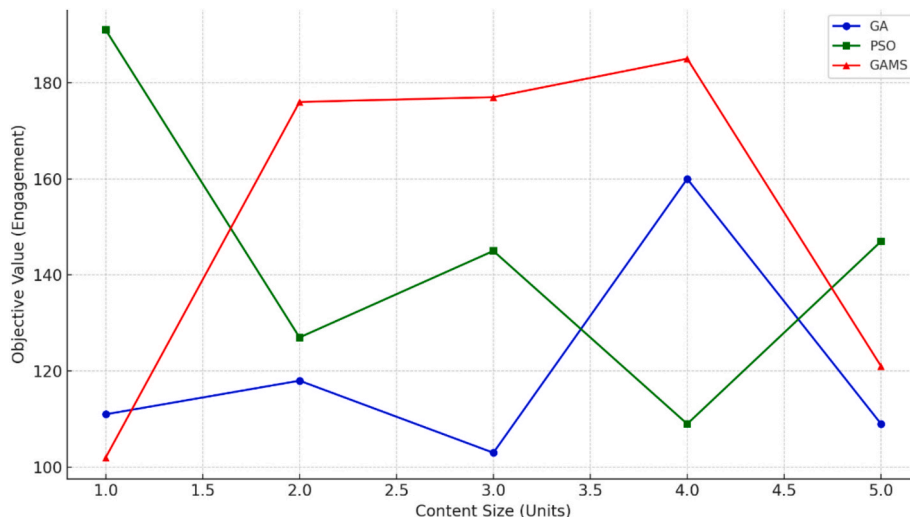


Fig. 5. Sensitivity analysis of algorithm performance to content volume.

is more important.

To examine these impacts, we first examined the capacity parameter of each platform, which is determined specifically for each platform. For example, Instagram has a capacity to accept 10 pieces of content, while Podcasts can only handle 6 pieces of content at a time. This difference in capacity causes the optimization algorithm to focus most of its attention on platforms with less capacity to maximize the utilization of content allocation to platforms.

Analyzing platform capacity helps us to make optimal content allocation considering the limited capacity of the platforms. This analysis is especially important for platforms with lower capacity, as these platforms cannot handle large volumes of content simultaneously and need to have more optimal allocation.

Fig. 6 examines the impact of changes in platform capacity on the value of the objective function (Engagement). In this figure, different platforms with different capacities (such as Instagram with a capacity of 10 and Podcast with a capacity of 6) are analyzed. As can be seen in the figure, platforms with higher capacities such as Instagram and YouTube have been able to attract a higher amount of engagement, because these platforms have the ability to distribute more content. On the other hand, platforms with more limited capacities such as TikTok and Podcast have a lower amount of engagement than high-capacity platforms, because they have less capacity to distribute content and cannot share the same amount of content that other platforms can manage.

After designing the optimization model and running it under baseline conditions, there is a need to evaluate the behavior of the model in the face of various changes; in other words, it should be determined how the model reacts and to what extent its results are stable when changing the main parameters or constraints. This study helps to better understand the generalizability of the model and provides a tool for predicting the behavior of the system in more realistic conditions. In this regard, four key components have been examined: platform capacity, distribution cost, interaction score, and budget constraints. These components, which are all part of the mathematical model formulation, have been changed in the form of independent scenarios so that the effect of each can be separated and analyzed.

Changing platform capacity is the first component to which the model's response is examined. In the basic model, the capacity of each platform is defined in a way that determines its operational limits in content allocation. Increasing or decreasing this capacity can affect the optimal allocation of content and the level of engagement achieved. For example, increasing the capacity of platforms such as Instagram and YouTube has caused a larger portion of the content produced to be allocated to these platforms, and as a result, the final level of engagement has increased. Conversely, when the capacity of high-traffic platforms was reduced, the model was forced to allocate content to less-engaged platforms, which resulted in a relative decrease in the overall engagement score.

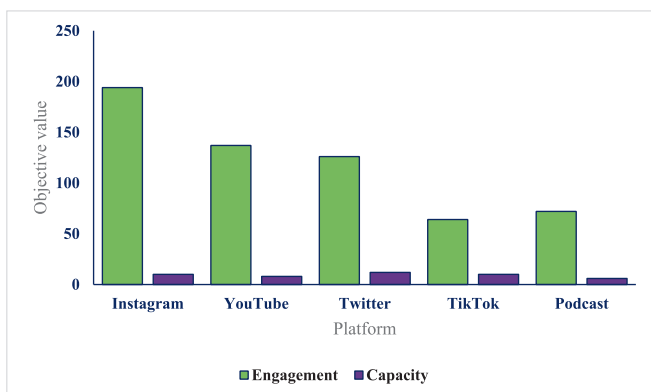


Fig. 6. Analysis of the impact of platform capacity on content allocation.

Table 5 presents a representation of the changes made to key parameters and their values under baseline and variable conditions. This table provides a basis for comparing and analyzing the model behavior.

The second analysis is dedicated to examining the impact of increasing distribution costs. This parameter is defined in the model as one of the main inputs to the cost function and directly affects the optimal mix of platforms. When the distribution cost range was increased, the model inevitably tended to use lower-cost platforms, even if these platforms generated less engagement. As a result, although the total cost did not decrease, the content allocation structure changed in such a way that it focused more on managing the budget constraint rather than maximizing engagement. This clearly shows that the model, under financial pressure, adjusts its priorities to avoid violating the constraints, even if it comes at the cost of a relative decrease in quality outputs such as engagement.

In the third scenario, the model was tested against changes in engagement scores. In this scenario, the base engagement scores of the platforms were increased, and the model was required to update its allocation decisions based on new information. The results of this analysis showed that if the engagement scores on some platforms increased, the model automatically became more inclined to allocate content to those platforms, even if their distribution costs were higher. In fact, the model's weighting of the engagement variable increased in such conditions, and the first objective function (maximizing engagement) dominated the cost function. This response is an indication of the model's flexibility in prioritizing multiple objectives and moving between them in response to environmental change.

Finally, a scenario with a reduced overall budget was considered to examine how the model would behave under severe financial constraints. In this situation, the budget was reduced from 1,200 to 1,000 units, and the model was forced to abandon some costly allocations. The model output in this scenario showed that although the overall interaction amount was reduced, an optimal allocation was still made within the budget. This indicates that the model has good adaptability to constrained conditions and can provide acceptable solutions despite external pressures.

Overall, the results of the model behavior analysis show that its mathematical structure is designed to be flexible to environmental and parametric changes. Whether in conditions of cost increases, budget constraints, or switching between different objectives, the model is able to produce consistent and logical responses that are operationally reliable and scientifically valid.

In addition to the numerical evaluation of the performance of the optimization model, a qualitative examination of the outputs also plays a decisive role in interpreting the results. A key aspect of this examination is the analysis of the final mix of content across platforms; in other words, determining how the model decided to distribute the content of the stories across media platforms in each scenario. This type of analysis is important because it goes beyond mere numerical optimization and reflects the decision-making logic of the model under real-world conditions and various constraints.

The results show that the optimal content allocation pattern is highly dependent on the environmental scenario conditions and the input parameters of the model. In the baseline scenario, the content allocation mix was almost balanced, such that platforms such as Instagram and

Table 5
Parameters used for scenario-based behavior.

Scenario Parameter	Baseline Value	Altered Scenario Value
Platform Capacity	IG: 10, YT: 8, TW: 12, TT: 10, PC: 6	IG: 6, YT: 6, TW: 10, TT: 8, PC: 4
Distribution Cost Range	[50, 200]	[80, 250]
Engagement Score Range	[30, 150]	[70, 200]
Total Budget	1200 USD	1000 USD

YouTube received the largest share due to their high engagement score and acceptable distribution cost. Twitter, as a medium-cost platform with adequate capacity, also had a relatively constant share in the content mix. However, in the scenario in which the engagement score of some platforms increased, we witnessed a noticeable change in the optimal mix; Instagram and YouTube received a higher share by increasing the weight in the engagement objective function, and platforms with weaker engagement were excluded from the mix. This behavior shows that the model automatically changes its prioritization in situations where the engagement objective prevails over other objectives. On the other hand, in the budget constraint scenario, the content mix changed again. In this case, the model tended to use platforms with low distribution cost more; In particular, Podcast and TikTok, although not performing well in terms of engagement, were given a higher position in the final allocation due to financial pressure. This pattern suggests that under economic pressure, the model accepts less engagement to meet the cost cap, but still tries to allocate in a way that keeps the total engagement at an acceptable level.

The optimal content mix in three different scenarios can be seen in Table 6. The values presented represent the relative percentage of content allocation to each platform out of the total story collection in each scenario.

As can be seen in Table 5, the allocation pattern in each scenario follows a logic consistent with the model structure. In the high engagement scenario, platforms with the potential to attract more engagement have found a superior position, and in the budget-constrained scenario, priority has shifted towards more economical platforms. An important point in all these analyses is the flexibility of the model in adapting to changing conditions and simultaneously meeting its multi-objective requirements; so that the content mix is not only numerically optimal, but also defensible from an implementation perspective and appropriate to the decision-making scenario.

7. Discussion and Conclusion

In this study, an intelligent framework was presented to optimize the distribution of transmedia narratives on different social platforms, in which the selection of the publication location of each part of the story was carried out simultaneously with the aim of maximizing user interaction and minimizing the distribution cost. The proposed model was designed with a multi-objective approach and realistic constraints such as platform capacity, publication costs, content technical compatibility score, assignable content volume, and budget ceiling, and was implemented and analyzed in a simulated environment.

Although the present analysis relies on simulated data, the model's structure is fully compatible with real-world datasets and can be readily calibrated using empirical values from actual transmedia or social media campaigns in future studies.

The results obtained from various analyses showed that the model has high internal coherence, logical behavior in the face of changing parameters, and significant flexibility. The analysis of platform capacity showed that networks with higher capacity, such as Instagram and YouTube, under normal conditions and without specific constraints, receive higher priority in content allocation and are more successful in attracting interaction. This result was further emphasized when the

optimal content mix analysis carefully examined the share of these platforms in different scenarios and observed that in scenarios with high engagement scores, this share even increases, while in budget-pressured situations, the priority intelligently shifts towards more economical platforms such as Podcast and TikTok.

On the other hand, the analysis of the model's behavior under different conditions showed that the proposed optimization system has a high ability to adapt to non-static and dynamic scenarios. The model exhibited realistic and adaptive behaviors in scenarios with changing capacity, cost, or financial constraints. For example, in conditions of increasing distribution costs, content allocation tended to be directed towards platforms with lower costs, even if their engagement scores were lower. In contrast, in scenarios of increasing engagement scores, the model tended to bear more costs, provided that it achieved higher engagement. Such behaviors indicate that the model is balancing two conflicting objective functions with appropriate accuracy and, depending on the circumstances, establishes a rational balance between them.

Sensitivity analysis also showed that changing the weights of the objective functions in the model leads to a significant change in the final content allocation. In other words, decision makers can easily control the orientation of the model based on their strategic goals (e.g., greater emphasis on financial savings or maximizing engagement). This feature transforms the model from a purely technical tool to a strategic tool that can be adapted to the goals of various campaigns.

From a computational perspective, the model was evaluated using exact solution in GAMS software as well as two metaheuristic algorithms, GA and PSO. Comparing the performance of these methods showed that exact solution with GAMS provides more optimal results in smaller problems, while metaheuristic algorithms are able to produce near-optimal solutions in a reasonable amount of time for larger problems with high variable volume. This provides an important practical advantage for the application of the model in real-world scales, where exact solution may not be possible.

In fact, the reason for the lower execution time of GA is due to the relative simplicity of the search structure. GA works mainly on the basis of selection, combination and mutation operators that traverse the search space with minimal matrix processing; while PSO requires continuous vector calculations and updating of positions at each iteration. In contrast, GAMS uses the exact solution of the system of mathematical equations and constraints, which, although leading to higher accuracy, significantly increases the calculation time. Therefore, GA is lighter in terms of computational complexity but has limitations in the ultimate accuracy compared to PSO and GAMS.

The most important innovations of this research include: designing a multi-objective hybrid model specific to transmedia, considering real operational factors such as budget constraints, content compatibility with the media, technical capacity, and distribution cost; using hybrid optimization methods including *meta*-heuristic algorithms and exact solution for comparative evaluation of model performance; and finally, multi-layered analyses including sensitivity analysis, model behavior analysis under different conditions, and optimal content combination analysis, which together provide a comprehensive picture of model behavior in real scenarios.

Finally, the findings of this research show that the proposed framework is not only valuable from a theoretical and algorithmic perspective, but also from a practical perspective, it can be used as a decision-making tool for content production teams, digital brands, and marketing campaign managers in the transmedia space.

In the current version, the proposed model is designed statically and its inputs are determined at the beginning of the decision-making process. However, one of the future development paths is to add learning and adaptability to changes in user behavior. This goal can be achieved by utilizing real-time user feedback (such as viewing rates, sharing, and emotional reactions in real time) and using machine learning models to dynamically update the interaction and cost parameters. In this way, the model will be able to automatically and continuously optimize the

Table 6
Optimal content allocation (%) across platforms in three key scenarios.

Platform	Baseline Scenario	High Engagement Scenario	Low Budget Scenario
Instagram	28 %	33 %	24 %
YouTube	25 %	30 %	18 %
Twitter	20 %	17 %	22 %
TikTok	15 %	12 %	20 %
Podcast	12 %	8 %	16 %

content distribution strategy and adapt itself to environmental changes and user preferences. This development path transforms the proposed framework from a static decision-making model to a dynamic and adaptive intelligent system for managing multimedia campaigns.

This framework has the ability to be extended to dynamic, event-driven, and data-driven multimedia environments and can be a basis for the development of intelligent decision-support systems in the field of digital content management.

Overall, the findings of this study showed that the use of hybrid approaches between exact optimization methods (such as GAMS) and metaheuristics (such as GA and PSO) can create an effective balance between two opposing goals, namely increasing audience engagement and reducing content distribution costs in multimedia narrative networks. A comparative analysis between the three algorithms revealed significant differences in terms of accuracy, computational time, and response diversity, indicating the importance of choosing the optimal method appropriate to the type of problem and its constraints.

The results of this study showed that the use of multi-algorithmic and hybrid approaches, in which the mathematical accuracy of exact methods is combined with the global search flexibility of metaheuristic algorithms, can significantly improve the efficiency of the decision-making process in complex and multi-objective problems. However, this study also faces some limitations; Including the model's reliance on synthetic data instead of real data and the failure to consider the temporal and behavioral dynamics of users in the optimization structure, which can affect the degree to which the model adapts to real conditions.

In line with future developments, it is suggested that the proposed model be extended towards Real-Time Adaptive Optimization, so that by utilizing machine learning and reinforcement learning technologies, it is possible to automatically adjust the parameters of multimedia campaigns based on real-time feedback from the audience. Also, validating the model with real industrial data and analyzing dynamic scenarios are future research directions that can help increase the accuracy, stability, and practical applicability of the proposed framework.

CRedit authorship contribution statement

Hamed Nozari: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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